

Chat With Energy Documents(RAG System)

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI - 13

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Problem Statement & Objectives

1. Problem Statement :

Energy sector organizations generate large volumes of technical documents such as operational manuals, turbine performance reports, grid compliance documents, sustainability reports, and maintenance logs. These documents contain critical information required for decision-making, troubleshooting, regulatory compliance, and operational optimization.

However, retrieving specific and contextually relevant information from such documents is difficult due to:

- Large unstructured PDF files
- Lack of semantic search capabilities
- Manual keyword-based searching
- Time-consuming document analysis
- Risk of misinterpretation
- Absence of AI-powered contextual understanding

Traditional document management systems rely on basic keyword search, which fails to understand semantic meaning or contextual relationships between terms. This results in inefficient workflows and delayed operational decisions.

Therefore, there is a need for a secure, scalable, and intelligent system that can:

- Process Energy domain documents
- Perform semantic search
- Retrieve context-aware information
- Generate grounded AI responses
- Provide accurate source references

The proposed solution is a production-ready Retrieval-Augmented Generation (RAG) system designed specifically for Energy Document Intelligence.

2. Project Objectives :

The primary objectives of this project are:

1. To design and implement a production-ready RAG-based system for Energy domain documents.

2. To enable secure PDF document upload and automated text processing.
3. To generate semantic embeddings for efficient contextual retrieval.
4. To store and index embeddings using a high-performance vector database (FAISS).
5. To retrieve the top relevant document chunks using cosine similarity search.
6. To generate grounded responses using GPT-based LLM models.
7. To ensure hallucination-free AI responses by restricting answers strictly to retrieved context.
8. To provide source traceability including document name and page number.
9. To implement JWT-based authentication for secure access.
10. To deploy the system in a production-ready Docker environment.

3. Scope of the Project :

The scope of this project includes:

- Uploading and validating Energy domain PDF documents.
- Extracting and cleaning text from uploaded PDFs.
- Splitting text into overlapping chunks (800 tokens with 150 overlap).
- Generating embeddings using OpenAI embedding models.
- Storing embeddings in FAISS vector database.
- Storing metadata in SQLite using SQLAlchemy ORM.
- Implementing semantic similarity search for retrieval.
- Generating context-aware answers using GPT-4o-mini.
- Providing structured JSON responses with source references.
- Implementing secure authentication and logging mechanisms.

Out of Scope:

- Direct integration with industrial IoT hardware
- Real-time control of manufacturing systems
- Multi-tenant enterprise deployment
- Large-scale distributed cloud architecture

Proposed Solution :

The proposed solution is a modular, scalable Retrieval-Augmented Generation (RAG) system that enables intelligent querying of Energy documents.

The system combines:

- Vector-based semantic retrieval
- Context-aware LLM generation
- Secure backend architecture
- Production-ready deployment

It ensures that responses are strictly grounded in the uploaded documents.

1. Key features :

- Secure PDF upload (20MB limit)
- Automatic text extraction and cleaning
- Custom recursive chunking (800 tokens, 150 overlap)
- Batch embedding generation
- FAISS-based semantic vector search
- Top-5 contextual retrieval
- Strict grounded prompt enforcement
- Hallucination prevention
- Source citation (document name + page number)
- JWT-based authentication
- Structured logging and monitoring
- Docker-based deployment

2. Overall Architecture / Workflow :

Document Upload Workflow -

1. User uploads PDF.
2. File validation is performed.
3. Text extraction using pdfplumber.

4. Text cleaning and preprocessing.
5. Recursive chunking applied.
6. Embeddings generated using OpenAI.
7. Embeddings stored in FAISS.
8. Metadata stored in SQLite.

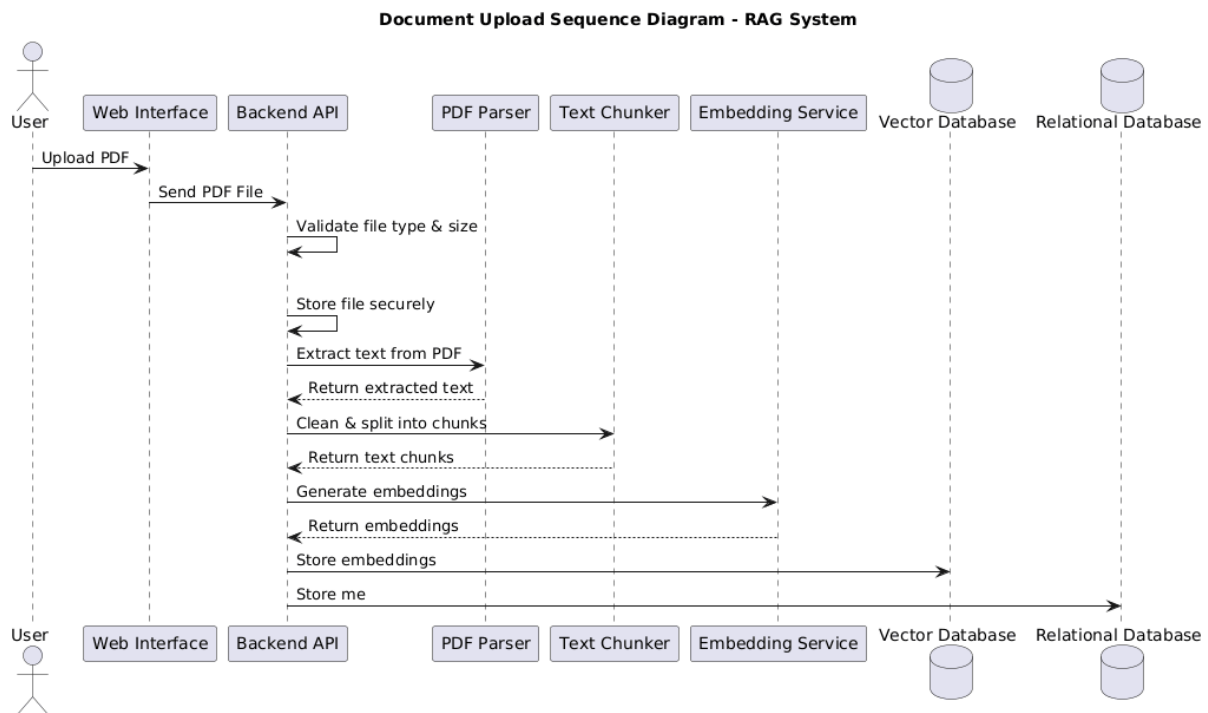


Fig 1: Document Upload Workflow

Question-Answer Workflow -

1. User submits authenticated query.
2. Query converted into embedding.
3. Top-5 similar chunks retrieved via cosine similarity.
4. Strict grounded prompt constructed.
5. Context + question sent to GPT-4o-mini.
6. AI generates response.
7. Response returned with document references.

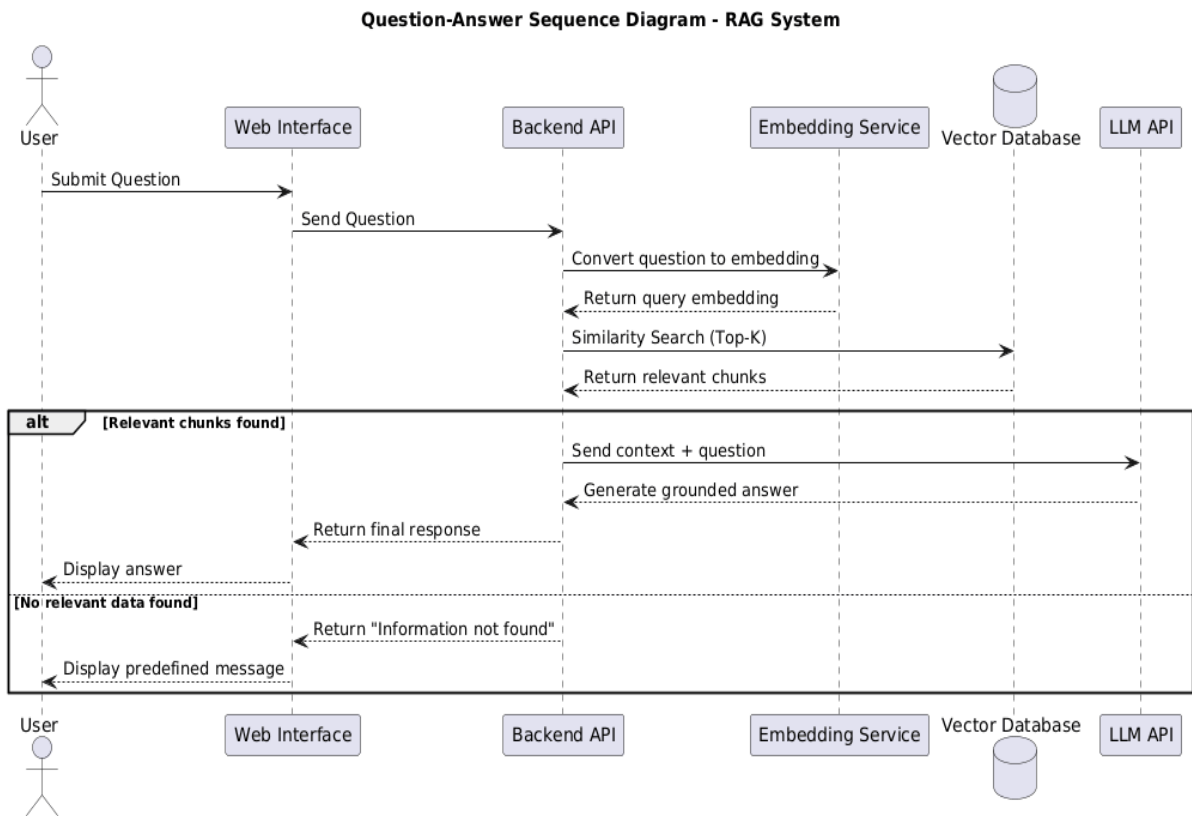


Fig 2: Question Answer Workflow

The system follows a clean layered architecture:

- API Layer (FastAPI)
- Service Layer (Business Logic)
- Repository Layer (Vector + Metadata Storage)
- Core Layer (Config + Security)

3. Tools & Technologies Used :

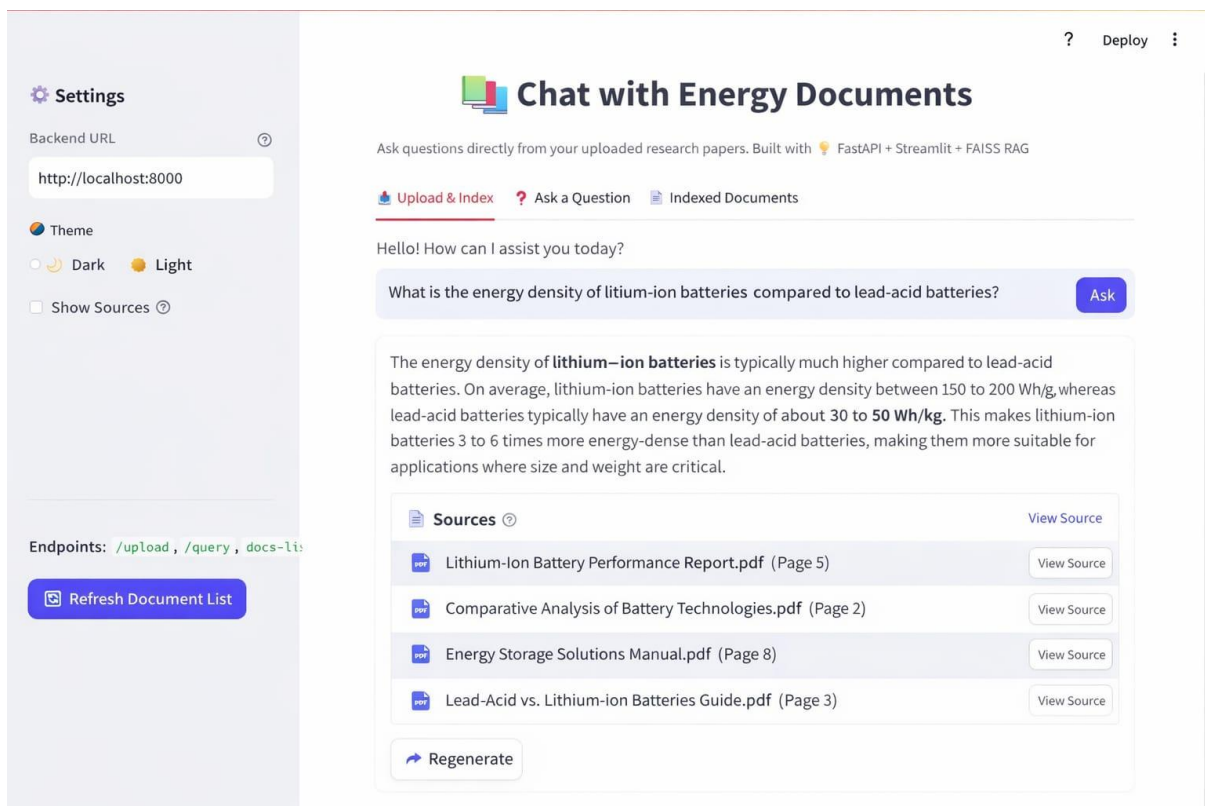
Component	Technology
Backend Framework	FastAPI
Language	Python 3.11
Embeddings	OpenAI text-embedding-3-large
LLM	OpenAI GPT-4o-mini
Vector Database	FAISS (local, extendable to Pinecone)

ORM	SQLAlchemy
Database	SQLite
PDF Processing	pdfplumber
Authentication	JWT
Configuration	Pydantic BaseSettings
Deployment	Docker
Logging	Python logging module

Results & Output

1. Screenshots / outputs

- Successful PDF upload confirmation
- Indexed document logs
- Chat query interface
- JSON response with answer and source references
- Performance logs showing embedding and retrieval time



The screenshot displays the 'Chat with Energy Documents' web application. On the left is a 'Settings' sidebar with options for Backend URL (http://localhost:8000), Theme (Dark/Light), and Show Sources. The main area features a chat interface with a title 'Chat with Energy Documents' and a subtitle 'Ask questions directly from your uploaded research papers. Built with FastAPI + Streamlit + FAISS RAG'. It includes tabs for 'Upload & Index', 'Ask a Question', and 'Indexed Documents'. A chat message asks 'What is the energy density of lithium-ion batteries compared to lead-acid batteries?', and the response states that lithium-ion batteries have a higher energy density (150-200 Wh/kg) compared to lead-acid batteries (30-50 Wh/kg). Below the response is a 'Sources' section listing four PDF documents with 'View Source' links: 'Lithium-Ion Battery Performance Report.pdf (Page 5)', 'Comparative Analysis of Battery Technologies.pdf (Page 2)', 'Energy Storage Solutions Manual.pdf (Page 8)', and 'Lead-Acid vs. Lithium-ion Batteries Guide.pdf (Page 3)'. A 'Regenerate' button is also present.

Fig. 3 Front page

2. Reports / dashboards / models

- Indexed chunk metadata table (SQLite)
- FAISS vector index structure
- Log monitoring output
- API response samples
- Retrieval similarity results

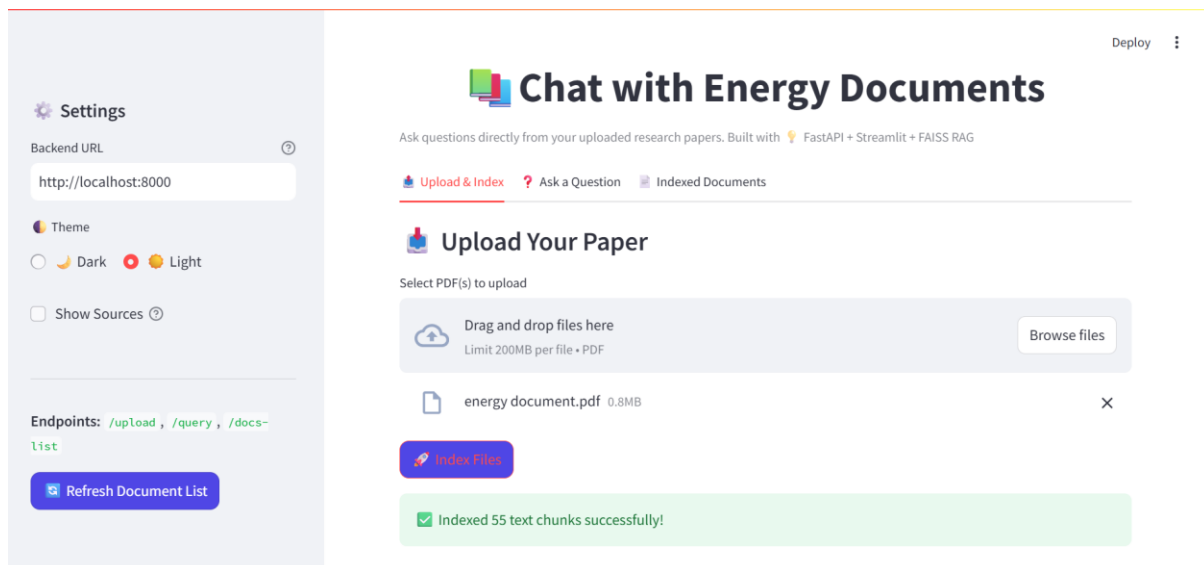


Fig. 4 : Indexed chunk from paper

3. Key outcomes

- Successfully implemented production-grade RAG architecture.
- Achieved semantic document retrieval using vector embeddings.
- Prevented hallucinated responses via strict prompt design.
- Enabled traceable AI responses with document references.
- Ensured secure access through JWT authentication.
- Delivered scalable and modular backend architecture.

Conclusion

The Energy Document Intelligence System successfully demonstrates the practical implementation of Retrieval-Augmented Generation (RAG) in a production-ready backend environment.

By integrating semantic embeddings, vector similarity search, and GPT-based language models, the system enables accurate, context-aware querying of Energy domain documents. The modular architecture, secure authentication, and Docker-based deployment ensure scalability and maintainability.

This project strengthened understanding in:

- Backend API design
- AI system integration
- Vector database implementation
- Prompt engineering
- Secure production deployment

Overall, the system provides a robust and scalable foundation for enterprise-level document intelligence solutions.

Future Scope & Enhancements

- Integration with Pinecone for distributed vector storage
- Hybrid search (BM25 + Vector search)
- Re-ranking models for improved retrieval accuracy
- Conversational memory support
- Multi-document and multi-user versioning
- Role-based access control
- Cloud deployment (AWS / GCP)
- Real-time monitoring dashboard
- Integration with enterprise document management systems